



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION FEBRUARY 2025

Course : EC3114 (Software Modelling and Analysis) Time: 2.00 pm – 5.00 pm (3 hours)
Lecturer : Mr. Izrulfizal Saufihamizal bin Ibrahim Date : 20 May 2025

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 2 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1.
 - a. What is the purpose of using software modeling, and how does it improve the software development process? **(4 MARKS)**
 - b. Explain what functional and non-functional requirements are in software engineering, and highlight how they differ from each other. **(4 MARKS)**
 - c. Why is a use case diagram important in software modeling? Provide an example to support your explanation. **(5 MARKS)**
 - d. Describe the various types of UML diagrams used in software modeling. For each diagram, explain its purpose and give an example of how it could be applied in a real-world situation. **(12 MARKS)**
2.
 - a. Explain the steps in the requirements gathering process and discuss why collecting accurate requirements early in software development is essential. **(7 MARKS)**
 - b. Define a sequence diagram and explain how it differs from a collaboration diagram. Use an example scenario to support your explanation. **(8 MARKS)**
 - c. What is Object-Oriented Analysis and Design (OOAD)? Describe how its principles can be used to create a strong and effective software architecture. **(10 MARKS)**

3. You are tasked with designing a class diagram for a simple Online Food Ordering System. The system should include functionalities for managing menus, customers, and orders. Your class diagram should include at least three classes, each with clearly defined attributes, methods, and relationships such as associations and generalizations. Provide explanations for your design choices and any assumptions you make. **(25 MARKS)**
4. Create an activity diagram to model the process of online food ordering through a food delivery application. The process should include activities such as browsing the menu, selecting items, placing an order, payment processing, and receiving an order confirmation. Clearly label all activities, decision points, and control flows. Additionally, explain how this activity diagram can assist in evaluating and enhancing the application's workflow. **(25 MARKS)**

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